

Please check the examination details below before entering your candidate information

Candidate surname	Other names
USMAN	Grade A* Scholar
Centre Number [9][5][1][0][2]	Candidate Number [0][0][4][4] Mentora Academy Internal Candidate Ref: MEN-EDX-2026-U4-W04

Pearson Edexcel International Advanced Level

Chemistry

International Advanced Level

UNIT 4: Rates, Equilibria and Further Organic Chemistry

WEEK 4 ASSESSMENT: TOPIC 11 SYNOPTIC & TOPIC 12A ENTROPY

Assessment Paper:	WCH14/01 (Week 4 In-Depth Mastery)
Time Allowed:	2 hours 30 minutes
Total Marks:	150 Marks
Topics Examined:	Topic 11 Kinetics Grand Synthesis (11A.1–11A.6) Topic 12A.1: Introduction to Entropy (S, Disorder, Molar Entropies) Topic 12A.2: Total Entropy (Surroundings, Feasibility Criteria)

INSTRUCTIONS TO CANDIDATES

- Use black ink or ball-point pen. Fill in the boxes at the top of this page.
- Answer ALL questions in Section A, Section B, and Section C.
- In Section A (Questions 1 to 30), select your answer by placing a clear cross in the box [X].
- In Section B and C, show all working in calculation questions. Include appropriate units where required.
- Questions marked with an asterisk (*) are those where your quality of written communication and logical sequence of structure will be assessed.
- A Periodic Table and physical constants are printed on the back page of this booklet.

INFORMATION FOR CANDIDATES

- The total mark for this paper is 150.
- Section A: 30 marks; Section B: 90 marks; Section C: 30 marks.
- Full worked confidential teacher mark scheme is provided at the rear for post-assessment diagnostic.

SECTION A

Answer ALL questions. For each question, make a selection by putting a cross in one box [X]. If you change your mind, put a line through the box and then mark your new choice with a cross.

1 Which of the following represents the correct SI units for standard molar entropy, S° ?

- ☐ **A** kJ mol^{-1}
☐ **B** J K^{-1}
☐ **C** $\text{J mol}^{-1} \text{K}^{-1}$
☐ **D** $\text{kJ mol}^{-1} \text{K}^{-1}$

(Total for Question 1 = 1 mark)

2 Which substance has the highest standard molar entropy at 298 K?

- ☐ **A** Diamond, C(s)
☐ **B** Liquid water, $\text{H}_2\text{O(l)}$
☐ **C** Graphite, C(s)
☐ **D** Steam, $\text{H}_2\text{O(g)}$

(Total for Question 2 = 1 mark)

3 Consider the reaction: $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$. Which statement correctly predicts and explains the sign of S_{system} ?

- ☐ **A** Positive, because one mole of gas reactant produces two moles of gas products, increasing disorder
☐ **B** Negative, because heat is absorbed during the endothermic bond dissociation
☐ **C** Positive, because nitrogen dioxide is a brown gas with higher kinetic energy
☐ **D** Negative, because the total number of nitrogen atoms remains constant

(Total for Question 3 = 1 mark)

4 For the industrial synthesis of ammonia: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$, what is the sign of S_{system} and the physical rationale?

- ☐ A Positive, because ammonia molecules are polar and possess dipole-dipole interactions
- ☐ B Negative, because 4 moles of gaseous reactants form 2 moles of gaseous product
- ☐ C Positive, because energy is released to the surroundings during this exothermic reaction
- ☐ D Zero, because all reacting species and products remain in the gas phase

(Total for Question 4 = 1 mark)

5 When solid magnesium sulfate, $\text{MgSO}_4(\text{s})$, dissolves in water, the entropy change of the system, S_{system} , is unexpectedly negative. What is the most plausible explanation for this observation?

- ☐ A Dissolving an ionic solid always decreases the entropy of the system
- ☐ B The hydration of Mg^{2+} and SO_4^{2-} ions causes a high degree of ordering of surrounding water molecules
- ☐ C The magnesium sulfate precipitates out of solution as an insoluble hydrate
- ☐ D The lattice enthalpy of magnesium sulfate is greater than its enthalpy of hydration

(Total for Question 5 = 1 mark)

6 Given standard molar entropies at 298 K:

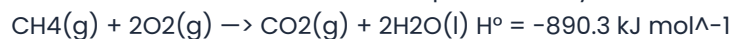
$S^\circ[\text{CaCO}_3(\text{s})] = 92.9 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\circ[\text{CaO}(\text{s})] = 38.2 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\circ[\text{CO}_2(\text{g})] = 213.6 \text{ J K}^{-1} \text{ mol}^{-1}$.

What is the value of S°_{system} for the thermal decomposition: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$?

- ☐ A $-158.9 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $+158.9 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $+268.3 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $+344.7 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 6 = 1 mark)

7 The combustion of methane is represented by:



Given standard molar entropies:

$$S^\circ[\text{CH}_4(\text{g})] = 186.2, S^\circ[\text{O}_2(\text{g})] = 205.0, S^\circ[\text{CO}_2(\text{g})] = 213.6, S^\circ[\text{H}_2\text{O}(\text{l})] = 69.9 \text{ J K}^{-1} \text{ mol}^{-1}.$$

What is the value of S°_{system} for this combustion at 298 K?

- ☐ A $-242.8 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $+242.8 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $-107.7 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $+107.7 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 7 = 1 mark)

8 Which mathematical expression correctly defines the entropy change of the surroundings, $S_{\text{surroundings}}$?

- ☐ A $S_{\text{surroundings}} = +H / T$
- ☐ B $S_{\text{surroundings}} = -H / T$
- ☐ C $S_{\text{surroundings}} = -T / H$
- ☐ D $S_{\text{surroundings}} = S_{\text{system}} - (H / T)$

(Total for Question 8 = 1 mark)

9 A reaction has an enthalpy change $H = -286 \text{ kJ mol}^{-1}$. What is the value of $S_{\text{surroundings}}$ at 298 K?

- ☐ A $-0.960 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $+0.960 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $-959.7 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $+959.7 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 9 = 1 mark)

10 An endothermic reaction absorbs 178 kJ mol^{-1} of heat at 500 K . What is the entropy change of the surroundings, $S_{\text{surroundings}}$?

- ☐ A $+356 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $-356 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $+0.356 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $-0.356 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 10 = 1 mark)

11 What is the fundamental thermodynamic criterion for a reaction to be spontaneous (feasible) under constant temperature and pressure?

- ☐ A $S_{\text{system}} > 0$
- ☐ B $S_{\text{surroundings}} > 0$
- ☐ C $S_{\text{total}} > 0$
- ☐ D $H < 0$

(Total for Question 11 = 1 mark)

12 A reaction is exothermic ($H < 0$) and results in a decrease in system entropy ($S_{\text{system}} < 0$). How does feasibility change as temperature increases?

- ☐ A Feasible at high temperatures, non-feasible at low temperatures
- ☐ B Feasible at low temperatures, non-feasible at high temperatures
- ☐ C Feasible at all temperatures
- ☐ D Non-feasible at all temperatures

(Total for Question 12 = 1 mark)

13 A reaction is endothermic ($H > 0$) and has a positive entropy change of the system ($S_{\text{system}} > 0$). At which temperatures is this reaction feasible?

- ☐ A Only below a certain crossover temperature
- ☐ B Only above a certain crossover temperature
- ☐ C At all temperatures
- ☐ D It can never be feasible at any temperature

(Total for Question 13 = 1 mark)

14 Under which set of thermodynamic parameters is a chemical reaction feasible at ALL temperatures?

- ☐ A H is positive and S_{system} is positive
- ☐ B H is negative and S_{system} is negative
- ☐ C H is negative and S_{system} is positive
- ☐ D H is positive and S_{system} is negative

(Total for Question 14 = 1 mark)

15 Under which set of thermodynamic conditions can a chemical reaction NEVER be feasible at any temperature?

- ☐ A H is positive and S_{system} is positive
- ☐ B H is negative and S_{system} is negative
- ☐ C H is negative and S_{system} is positive
- ☐ D H is positive and S_{system} is negative

(Total for Question 15 = 1 mark)

16 Synoptic Kinetics: Initial rate data for the reaction $2A + B \rightarrow C$ show that doubling $[A]$ while keeping $[B]$ constant quadruples the rate. Doubling $[B]$ while keeping $[A]$ constant doubles the rate. What is the overall order of reaction?

- ☐ A 1
- ☐ B 2
- ☐ C 3
- ☐ D 4

(Total for Question 16 = 1 mark)

17 Synoptic Kinetics: A reaction has the rate equation: $\text{rate} = k[X]^2[Y]$. What are the correct units of the rate constant, k ?

- ☐ A $\text{mol dm}^{-3} \text{s}^{-1}$
- ☐ B $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$
- ☐ C $\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$
- ☐ D s^{-1}

(Total for Question 17 = 1 mark)

18 How does an increase in temperature affect the rate constant k and the total entropy change S_{total} for an exothermic reaction with $S_{\text{system}} < 0$?

- ☐ A k increases; S_{total} increases
- ☐ B k increases; S_{total} decreases
- ☐ C k decreases; S_{total} increases
- ☐ D k decreases; S_{total} decreases

(Total for Question 18 = 1 mark)

19 Synoptic Kinetics: For the multistep reaction:

Step 1: $\text{NO}_2 + \text{NO}_2 \rightarrow \text{NO}_3 + \text{NO}$ (slow)

Step 2: $\text{NO}_3 + \text{CO} \rightarrow \text{NO}_2 + \text{CO}_2$ (fast)

Which is the experimentally deduced rate equation?

- ☐ A rate = $k[\text{NO}_2][\text{CO}]$
- ☐ B rate = $k[\text{NO}_2]^2$
- ☐ C rate = $k[\text{NO}_3][\text{CO}]$
- ☐ D rate = $k[\text{NO}_2]^2[\text{CO}]$

(Total for Question 19 = 1 mark)

20 Synoptic Kinetics: In an Arrhenius plot of $\ln k$ against $1/T$, the line has a gradient of $-9.62 \times 10^3 \text{ K}$. What is the activation energy, E_a , of the reaction? ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- ☐ A +1.16 kJ mol^{-1}
- ☐ B +79.98 kJ mol^{-1}
- ☐ C -79.98 kJ mol^{-1}
- ☐ D +1157 kJ mol^{-1}

(Total for Question 20 = 1 mark)

21 Graphite has a standard molar entropy $S^\circ = 5.7 \text{ J K}^{-1} \text{ mol}^{-1}$, whereas diamond has $S^\circ = 2.4 \text{ J K}^{-1} \text{ mol}^{-1}$. What is the main structural reason for this difference?

- ☐ A Diamond has weaker London dispersion forces between layers
- ☐ B Graphite has a lower density and its layers have vibrational freedom to slide
- ☐ C Diamond contains ionic impurities that restrict microstates
- ☐ D Graphite undergoes rapid sublimation at room temperature

(Total for Question 21 = 1 mark)

22 For a reaction at 298 K, $S_{\text{system}} = -198.5 \text{ J K}^{-1} \text{ mol}^{-1}$ and $H = -92.2 \text{ kJ mol}^{-1}$. What is the value of S_{total} ?

- ☐ A $-507.9 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $+110.9 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $-110.9 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $+507.9 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 22 = 1 mark)

23 For the vaporisation of water: $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$, $H_{\text{vap}} = +40.7 \text{ kJ mol}^{-1}$ and $S_{\text{vap}} = +109.1 \text{ J K}^{-1} \text{ mol}^{-1}$. What is the boiling temperature of water at 1 atm pressure deduced from these thermodynamic data?

- ☐ A 373.0 K
- ☐ B 2.68 K
- ☐ C 0.373 K
- ☐ D 443.6 K

(Total for Question 23 = 1 mark)

24 A mixture of methane and oxygen is stored at 298 K. The reaction $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ has a very large positive S_{total} ($+2745 \text{ J K}^{-1} \text{ mol}^{-1}$), yet no reaction is observed for decades. Why?

- ☐ A The reaction is thermodynamically non-feasible at room temperature
- ☐ B The activation energy is very high, so the reaction is kinetically inert
- ☐ C Methane molecules do not possess any kinetic energy at 298 K
- ☐ D The products undergo rapid reverse dissociation back to reactants

(Total for Question 24 = 1 mark)

25 Trouton's rule states that the entropy of vaporisation for most normal liquids is approximately $+88 \text{ J K}^{-1} \text{ mol}^{-1}$. Why is the entropy of vaporisation of water ($S_{\text{vap}} = +109 \text{ J K}^{-1} \text{ mol}^{-1}$) significantly higher than this value?

- ☐ A Water vapour molecules are diatomic and have higher rotational freedom
- ☐ B Extensive hydrogen bonding in liquid water makes it unusually ordered
- ☐ C Water decomposes into hydrogen and oxygen gas upon boiling
- ☐ D The covalent O-H bonds break completely during vaporisation

(Total for Question 25 = 1 mark)

26 Synoptic Kinetics: A radioactive isotope decays by a first-order process with rate constant $k = 2.82 \times 10^{-3} \text{ s}^{-1}$. What is the half-life of this isotope?

- ☐ A 123 s
- ☐ B 246 s
- ☐ C 354 s
- ☐ D 709 s

(Total for Question 26 = 1 mark)

27 Synoptic Kinetics: When the temperature of a reaction mixture is increased by 10 K (e.g. from 300 K to 310 K), the rate of reaction approximately doubles. What is the primary reason for this dramatic increase?

- ☐ A The average kinetic energy of the molecules doubles
- ☐ B The collision frequency between reactant molecules doubles
- ☐ C The fraction of molecules with energy $E \geq E_a$ increases significantly
- ☐ D The activation energy of the reaction is halved

(Total for Question 27 = 1 mark)

28 When ammonium nitrate, $\text{NH}_4\text{NO}_3(\text{s})$, dissolves in water at 298 K, the temperature drops noticeably. Which row correctly gives the signs of H , S_{system} , and $S_{\text{surroundings}}$ for this process?

- ☐ A H : Positive (+) | S_{system} : Positive (+) | $S_{\text{surroundings}}$: Negative (-)
- ☐ B H : Negative (-) | S_{system} : Positive (+) | $S_{\text{surroundings}}$: Positive (+)
- ☐ C H : Positive (+) | S_{system} : Negative (-) | $S_{\text{surroundings}}$: Positive (+)
- ☐ D H : Negative (-) | S_{system} : Negative (-) | $S_{\text{surroundings}}$: Negative (-)

(Total for Question 28 = 1 mark)

29 Liquid water freezes to ice at 268 K (-5°C). Given that freezing is exothermic ($H = -6.01 \text{ kJ mol}^{-1}$) and $S_{\text{system}} = -22.0 \text{ J K}^{-1} \text{ mol}^{-1}$, what is the value of S_{total} at 268 K?

- ☐ A $-0.43 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $+0.43 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $+44.4 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $-44.4 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 29 = 1 mark)

30 Why is the standard entropy of hydration of the magnesium ion, $\text{Mg}^{2+}(\text{aq})$ ($-268 \text{ J K}^{-1} \text{ mol}^{-1}$), significantly more negative than that of the barium ion, $\text{Ba}^{2+}(\text{aq})$ ($-158 \text{ J K}^{-1} \text{ mol}^{-1}$)?

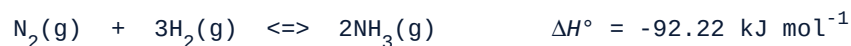
- ☐ A Mg^{2+} has a smaller ionic radius and higher charge density, attracting and immobilising water molecules more tightly
- ☐ B Ba^{2+} forms covalent bonds with surrounding water molecules
- ☐ C Mg^{2+} hydrolyses water molecules into hydrogen ions and magnesium oxide
- ☐ D Ba^{2+} has more electrons and therefore causes greater dispersion of solvent energy

(Total for Question 30 = 1 mark)

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

31 Ammonia is synthesised industrially from nitrogen and hydrogen via the Haber process:



Standard molar entropies, S° , at 298 K are provided in the table below:

Substance	$\text{N}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{NH}_3(\text{g})$
$S^\circ / \text{J K}^{-1} \text{mol}^{-1}$	191.6	130.6	192.3

(a) Explain why the standard molar entropy of ammonia ($192.3 \text{ J K}^{-1} \text{mol}^{-1}$) is greater than that of hydrogen ($130.6 \text{ J K}^{-1} \text{mol}^{-1}$), referring to molecular complexity and microstates.

(2)

.....

.....

.....

.....

(b) Calculate the standard entropy change of the system, S°_{system} , for the forward reaction in $\text{J K}^{-1} \text{mol}^{-1}$. Include a sign in your answer.

(3)

.....

.....

.....

.....

.....

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(c) Calculate the entropy change of the surroundings, $S_{\text{surroundings}}$, in $\text{J K}^{-1} \text{mol}^{-1}$:

(i) at 298 K

(ii) at 700 K (typical industrial operating temperature).

Assume that H° remains independent of temperature.

(4)

$S_{\text{surroundings}}$ (298 K) = $\text{J K}^{-1} \text{mol}^{-1}$ $S_{\text{surroundings}}$ (700 K) = $\text{J K}^{-1} \text{mol}^{-1}$

(d) Calculate the total entropy change, S_{total} , at 298 K and at 700 K. Hence deduce whether the forward reaction is thermodynamically feasible at each temperature.

(4)

S_{total} (298 K) = Feasible?
 S_{total} (700 K) = Feasible?

(e) Calculate the crossover temperature, $T_{\text{crossover}}$, in Kelvin, above which the forward synthesis of ammonia ceases to be thermodynamically feasible.

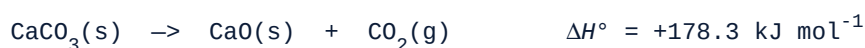
Explain why chemical engineers operate the Haber reactor at 700 K (approx 430 °C), despite the thermodynamic penalty on feasibility.

(5)

$T_{\text{crossover}}$ = K

(Total for Question 31 = 18 marks)

32 Limestone undergoes thermal decomposition in industrial lime kilns according to the equation:



Standard molar entropies at 298 K are shown below:

Compound	$\text{CaCO}_3(\text{s})$	$\text{CaO}(\text{s})$	$\text{CO}_2(\text{g})$
$S^\circ / \text{J K}^{-1} \text{mol}^{-1}$	92.9	38.2	213.6

(a) Explain, in terms of arrangement of particles and quanta of energy, why the entropy change of the system, S°_{system} , is positive for this reaction.

(2)

.....

.....

.....

.....

(b) Calculate S°_{system} for this thermal decomposition in $\text{J K}^{-1} \text{mol}^{-1}$.

(3)

.....

.....

.....

.....

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(c) Calculate $S_{\text{surroundings}}$ and S_{total} for this reaction at 298 K.

State why limestone blocks used in ancient buildings have persisted for thousands of years without decomposing.

(4)

.....

.....

.....

.....

.....

.....

.....

$S_{\text{total}} (298 \text{ K}) = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(d) Calculate the minimum temperature, in °C, required for the thermal decomposition of calcium carbonate to become thermodynamically feasible.

(4)

Minimum temperature = °C

(e) Magnesium carbonate, MgCO_3 , decomposes at a significantly lower temperature (approx 300 °C) than calcium carbonate, CaCO_3 (approx 840 °C).

Explain this difference in thermal stability with reference to:

- the ionic radii of Mg^{2+} and Ca^{2+}
- charge density and polarisation of the carbonate ion
- the relative lattice energies of the reactants and products.

(5)

(Total for Question 32 = 18 marks)

33 A student investigated the thermochemistry of dissolving ammonium nitrate in water for use in instant chemical cold packs:



A mass of 8.00 g of ammonium nitrate was dissolved in 50.0 cm³ of deionised water in an expanded polystyrene cup. The temperature of the water dropped by 12.4 °C.

(Assume the specific heat capacity of the solution is 4.18 J g⁻¹ K⁻¹ and the density of the solution is 1.00 g cm⁻³.

Molar mass of NH₄NO₃ = 80.0 g mol⁻¹).

(a) Calculate the enthalpy change of solution, H_{sol}[°], in kJ mol⁻¹. Include a sign in your answer. **(4)**

H_{sol}[°] = kJ mol⁻¹

(b) Literature data state that the standard entropy change of the system for this dissolution is

S_{system}[°] = +108.7 J K⁻¹ mol⁻¹.

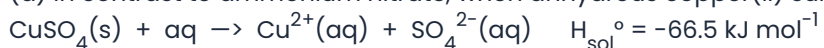
Using your value of H_{sol}[°] from (a), calculate S_{surroundings} and S_{total} for the dissolution at 298 K. **(4)**

S_{surroundings} = J K⁻¹ mol⁻¹ S_{total} = J K⁻¹ mol⁻¹

(c) Explain why this dissolution process occurs spontaneously at 298 K despite being endothermic, making explicit reference to the Second Law of Thermodynamics.

(3)

(d) In contrast to ammonium nitrate, when anhydrous copper(II) sulfate dissolves in water:



The entropy change of the system, $S^{\circ}_{\text{system}}$, is negative ($-156.0 \text{ J K}^{-1} \text{ mol}^{-1}$).

Explain why dissolving an ionic solid can result in a negative entropy change of the system.

(3)

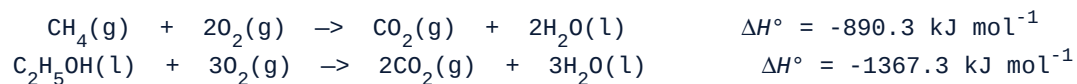
(e) Calculate the temperature, in $^{\circ}\text{C}$, above which the dissolution of anhydrous copper(II) sulfate ceases to be thermodynamically feasible. State one assumption made in your calculation.

(4)

Maximum temperature = $^{\circ}\text{C}$

(Total for Question 33 = 18 marks)

*34 Methane, CH_4 , and ethanol, $\text{C}_2\text{H}_5\text{OH}$, are widely utilised fuels. Their combustion reactions under standard conditions are represented by:



Thermodynamic data at 298 K are shown below:

Substance	$\text{CH}_4(\text{g})$	$\text{C}_2\text{H}_5\text{OH}(\text{l})$	$\text{O}_2(\text{g})$	$\text{CO}_2(\text{g})$	$\text{H}_2\text{O}(\text{l})$
$S^\circ / \text{J K}^{-1} \text{mol}^{-1}$	186.2	160.7	205.0	213.6	69.9

(a) For the combustion of methane at 298 K, calculate:

(i) S°_{system} in $\text{J K}^{-1} \text{mol}^{-1}$

(ii) $S_{\text{surroundings}}$ in $\text{J K}^{-1} \text{mol}^{-1}$

(iii) S_{total} in $\text{J K}^{-1} \text{mol}^{-1}$.

(5)

$S^\circ_{\text{system}} = \dots\dots\dots S_{\text{surroundings}} = \dots\dots\dots S_{\text{total}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

*(b) For the combustion of ethanol, $S^{\circ}_{\text{system}} = -138.1 \text{ J K}^{-1} \text{ mol}^{-1}$.

Compare and evaluate the thermodynamic feasibility and energy output of burning methane versus bioethanol.

In your answer, you should include:

- calculation of S_{total} for ethanol at 298 K
- an explanation of why both reactions have large positive S_{total} despite $S^{\circ}_{\text{system}}$ being negative
- the effect of an increase in combustion temperature on $S_{\text{surroundings}}$ and overall feasibility
- a comparison of the environmental impact (carbon neutrality and greenhouse warming potential).

(6)

(c) Natural gas pipelines transport methane and oxygen mixtures without explosion occurring at 298 K, despite S_{total} being overwhelmingly positive ($+2745 \text{ J K}^{-1} \text{ mol}^{-1}$).

Distinguish clearly between thermodynamic feasibility and kinetic stability, referring to collision theory and activation energy.

(4)

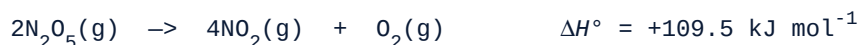
(d) Water is produced as a liquid in standard combustion reactions. Given that the enthalpy of vaporisation of water is $H_{\text{vap}} = +40.7 \text{ kJ mol}^{-1}$ at its normal boiling point (373 K), calculate the entropy of vaporisation, S_{vap} , of water in $\text{J K}^{-1} \text{ mol}^{-1}$.

(3)

.....
.....
.....
.....
.....
.....
 $S_{\text{vap}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(Total for Question 34 = 18 marks)

35 Dinitrogen pentoxide decomposes in the gas phase according to the stoichiometric equation:



The reaction is first order with respect to N_2O_5 . Experimental rate constants at two different temperatures are tabulated below:

Temperature T / K	298 K	338 K
Rate constant k / s ⁻¹	3.46×10^{-5}	4.87×10^{-3}

(a) Calculate the half-life, $t_{1/2}$, of the decomposition of N_2O_5 at 298 K in seconds and in hours.

Explain how a concentration-time graph can be used to confirm that the reaction is first order.

(4)

$t_{1/2} = \dots \text{ s} = \dots \text{ hours}$

(b) Use the two-point form of the Arrhenius equation:

$$\ln(k_2 / k_1) = (E_a / R) \times (1/T_1 - 1/T_2)$$

to calculate the activation energy, E_a , of the decomposition in kJ mol^{-1} . ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

(4)

$E_a = \dots \text{ kJ mol}^{-1}$

Standard molar entropies at 298 K: $S^\circ[\text{N}_2\text{O}_5(\text{g})] = 356.0$, $S^\circ[\text{NO}_2(\text{g})] = 240.0$, $S^\circ[\text{O}_2(\text{g})] = 205.0 \text{ J K}^{-1} \text{ mol}^{-1}$.

(c) Calculate the standard entropy change of the system, S°_{system} , for the decomposition in $\text{J K}^{-1} \text{ mol}^{-1}$.

(3)

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$

(d) Calculate $S_{\text{surroundings}}$ and S_{total} for this reaction at 298 K. Hence deduce whether the decomposition is thermodynamically feasible at room temperature.

(4)

$S_{\text{surroundings}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$ $S_{\text{total}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$

(e) When the temperature is raised to 350 K, N_2O_5 decomposes rapidly with evolution of brown NO_2 fumes.

Explain this rapid decomposition by evaluating BOTH:

- the kinetic effect (Maxwell-Boltzmann distribution and fraction of molecules with $E \geq E_a$)
- the thermodynamic feasibility effect (how an increase in temperature impacts $S_{\text{surroundings}}$ and S_{total}).

(3)

(Total for Question 35 = 18 marks)

SECTION C

Answer ALL questions. Write your answers in the spaces provided.

36 The transition to a hydrogen fuel economy depends heavily on the industrial production of hydrogen via the steam reforming of natural gas:



Standard molar entropies at 298 K are tabulated below:

Substance	$\text{CH}_4(\text{g})$	$\text{H}_2\text{O}(\text{g})$	$\text{CO}(\text{g})$	$\text{H}_2(\text{g})$
$S^\circ / \text{J K}^{-1} \text{mol}^{-1}$	186.2	188.7	197.6	130.6

(a) Calculate the standard entropy change of the system, S°_{system} , for the steam reforming reaction in $\text{J K}^{-1} \text{mol}^{-1}$. (3)

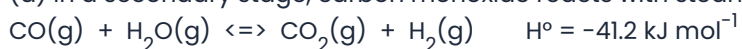
$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(b) Calculate the minimum temperature, in $^\circ\text{C}$, required for steam reforming of methane to become thermodynamically feasible ($S_{\text{total}} \geq 0$). (4)

Minimum temperature = $\dots\dots\dots ^\circ\text{C}$

- (c) Industrial reforming plants operate at 850 °C and 25 atm in the presence of a nickel catalyst.
- (i) Explain why high temperatures (850 °C) are chosen despite the massive energy costs, referring to S_{total} and equilibrium yield.
- (ii) Using Le Chatelier's principle, explain why operating at high pressure (25 atm) decreases the equilibrium yield of hydrogen.
- (iii) Suggest why high pressure is nevertheless used commercially in industrial plants. **(5)**

(d) In a secondary stage, carbon monoxide reacts with steam via the water-gas shift reaction:



Given that $S^\circ_{\text{system}} = -42.1 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate S_{total} at 500 K.

Explain why the water-gas shift converter must be operated at a much lower temperature (200 – 350 °C) than the primary reformer. **(3)**

$S_{\text{total}}(500 \text{ K}) = \dots\dots\dots \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 36 = 15 marks)

37 Biochemical macromolecules such as the enzyme lysozyme undergo reversible folding and unfolding in aqueous solution:

Unfolded (random coil) $\rightleftharpoons$ Folded (native globular conformation)

Calorimetric measurements of lysozyme folding at 298 K yield the following thermodynamic parameters:

- Enthalpy of folding: $H^\circ_{\text{fold}} = -280.0 \text{ kJ mol}^{-1}$
- Entropy of folding of the polypeptide system: $S^\circ_{\text{system}} = -752.0 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Explain why the folding of an extended, flexible polypeptide chain into a rigid, compact globular enzyme causes a large negative entropy change of the system ($S^\circ_{\text{system}} < 0$).

(2)

.....

.....

.....

.....

(b) Calculate $S_{\text{surroundings}}$ and S_{total} for the folding of lysozyme at 298 K (25 °C). Deduce whether the enzyme is thermodynamically stable in its folded native conformation at physiological room temperature.

(4)

.....

.....

.....

.....

.....

.....

.....

$S_{\text{surroundings}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$ $S_{\text{total}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$

(c) When heated, the enzyme undergoes thermal denaturation. Calculate the melting temperature, T_m , in $^{\circ}\text{C}$, at which the folded and unfolded forms exist in equal concentrations at equilibrium (where $S_{\text{total}} = 0$).

(4)

$T_m = \dots\dots\dots^{\circ}\text{C}$

(d) In aqueous solution, non-polar hydrophobic amino acid side chains (such as leucine and phenylalanine) are buried within the interior core of the folded protein.

Explain the 'hydrophobic effect' in terms of entropy:

- how water molecules arrange themselves around non-polar side chains in the unfolded state
- what happens to these water molecules when the protein folds
- why this release of water molecules provides an entropic driving force that helps overcome the conformational ordering of the polypeptide backbone.

(5)

(Total for Question 37 = 15 marks)

DATA SHEET: PERIODIC TABLE OF THE ELEMENTS & PHYSICAL CONSTANTS

Physical Constant	Symbol	Numerical Value & Units
Gas constant	R	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	L or N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Ionic product of water (298 K)	K_w	$1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
Specific heat capacity of water	c	$4.18 \text{ J g}^{-1} \text{ K}^{-1} = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$
Standard pressure	p°	$1.00 \times 10^5 \text{ Pa} = 100 \text{ kPa} = 1.00 \text{ bar}$
Standard temperature	T°	$298.15 \text{ K} = 25.0 \text{ }^\circ\text{C}$

Group →	1	2	3	4	5	6	7	0 / 8
Period 1	H 1.0							He 4.0
Period 2	Li 6.9	Be 9.0	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0	Ne 20.2
Period 3	Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5	Ar 39.9
Period 4	K 39.1	Ca 40.1	Ga 69.7	Ge 72.6	As 74.9	Se 79.0	Br 79.9	Kr 83.8
Period 5	Rb 85.5	Sr 87.6	In 114.8	Sn 118.7	Sb 121.8	Te 127.6	I 126.9	Xe 131.3
Period 6	Cs 132.9	Ba 137.3	Tl 204.4	Pb 207.2	Bi 209.0	Po [209]	At [210]	Rn [222]

CONFIDENTIAL TEACHER MARK SCHEME & EXAMINER TRAP COMMENTARY

Pearson Edexcel International Advanced Level Chemistry – Unit 4 (WCH14/01) – Week 4 Assessment (150 Marks)

Q	Correct Answer	Key Concept & Examiner Guidance	Mark
1	C	SI units of entropy are $\text{J K}^{-1} \text{mol}^{-1}$ (or $\text{J mol}^{-1} \text{K}^{-1}$). kJ is used for enthalpy.	1
2	D	Steam, $\text{H}_2\text{O}(\text{g})$, has gaseous state with rapid translational movement and highest disorder.	1
3	A	1 mol of gas reactant forms 2 mol of gas products: $n_{\text{gas}} = +1 > 0 \Rightarrow S_{\text{system}} > 0$.	1
4	B	4 mol gas ($1 \text{N}_2 + 3 \text{H}_2$) $\rightarrow$ 2 mol gas (2NH_3): $n_{\text{gas}} = -2 \Rightarrow S_{\text{system}}$ is negative.	1
5	B	High charge density of Mg^{2+} and SO_4^{2-} polarises and orders hydration shell water molecules.	1
6	B	$S^\circ = S^\circ(\text{CaO}) + S^\circ(\text{CO}_2) - S^\circ(\text{CaCO}_3) = 38.2 + 213.6 - 92.9 = +158.9 \text{ J K}^{-1} \text{mol}^{-1}$.	1
7	A	$S^\circ = [213.6 + 2(69.9)] - [186.2 + 2(205.0)] = 353.4 - 596.2 = -242.8 \text{ J K}^{-1} \text{mol}^{-1}$.	1
8	B	Definition of surroundings entropy: $S_{\text{surroundings}} = -H / T$.	1
9	D	$S_{\text{surr}} = -(-286\,000 \text{ J mol}^{-1}) / 298 \text{ K} = +959.7 \text{ J K}^{-1} \text{mol}^{-1}$.	1
10	B	$S_{\text{surr}} = -(+178\,000 \text{ J mol}^{-1}) / 500 \text{ K} = -356 \text{ J K}^{-1} \text{mol}^{-1}$.	1
11	C	Second Law of Thermodynamics: Feasibility requires $S_{\text{total}} > 0$.	1
12	B	Exothermic with $S_{\text{sys}} < 0$: S_{surr} is large positive at low T (feasible), but decreases as T rises.	1
13	B	Endothermic with $S_{\text{sys}} > 0$: $-H/T$ becomes less negative as T rises, so $S_{\text{total}} > 0$ at high T.	1
14	C	$H < 0$ ($S_{\text{surr}} > 0$) and $S_{\text{sys}} > 0 \Rightarrow S_{\text{total}} = S_{\text{sys}} + S_{\text{surr}}$ is positive at all T.	1
15	D	$H > 0$ ($S_{\text{surr}} < 0$) and $S_{\text{sys}} < 0 \Rightarrow S_{\text{total}}$ is negative at all T (never feasible).	1
16	C	Rate = $k[\text{A}]^2[\text{B}]^1 \Rightarrow$ overall order = $2 + 1 = 3$.	1
17	C	Units of k for 3rd order: $(\text{mol dm}^{-3} \text{s}^{-1}) / (\text{mol dm}^{-3})^3 = \text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$.	1
18	B	k always increases with T (Arrhenius); for exothermic, $S_{\text{surr}} = -H/T$ decreases $\Rightarrow S_{\text{total}}$ decreases.	1
19	B	Step 1 is slow RDS involving 2 molecules of $\text{NO}_2 \Rightarrow$ rate = $k[\text{NO}_2]^2$. CO is in fast Step 2.	1
20	B	Gradient = $-E_a / R \Rightarrow E_a = -(-9.62 \times 10^3) \times 8.314 = +79\,980 \text{ J mol}^{-1} = +79.98 \text{ kJ mol}^{-1}$.	1
21	B	Graphite has layer structure with weak inter-layer forces and more vibrational microstates.	1
22	B	$S_{\text{surr}} = -(-92\,200) / 298 = +309.4 \text{ J K}^{-1} \text{mol}^{-1}$; $S_{\text{total}} = -198.5 + 309.4 = +110.9 \text{ J K}^{-1} \text{mol}^{-1}$.	1
23	A	At boiling point $S_{\text{total}} = 0 \Rightarrow T = H_{\text{vap}} / S_{\text{vap}} = 40\,700 / 109.1 = 373.0 \text{ K}$.	1
24	B	High activation energy E_a creates a massive kinetic barrier; reaction is kinetically inert.	1
25	B	Liquid water contains hydrogen-bonded networks, giving it unusually low liquid entropy.	1
26	B	For first-order decay: $t_{1/2} = \ln 2 / k = 0.69315 / 0.00282 = 245.8 \text{ s} \approx 246 \text{ s}$.	1
27	C	Higher T shifts Maxwell-Boltzmann distribution, exponentially increasing molecules with $E \geq E_a$.	1
28	A	Dissolving absorbs heat ($H > 0$), increases disorder ($S_{\text{sys}} > 0$), takes heat from surroundings ($S_{\text{surr}} < 0$).	1
29	B	$S_{\text{surr}} = -(-6010 \text{ J mol}^{-1}) / 268 \text{ K} = +22.43$; $S_{\text{total}} = -22.0 + 22.43 = +0.43 \text{ J K}^{-1} \text{mol}^{-1}$ (spontaneous!).	1

30	A	Small Mg^{2+} has high charge density, immobilising hydration water molecules into ordered shell.	1
----	---	--	---

Question 31: Haber Process Thermodynamics & Crossover Feasibility (18 Marks)

(a) [2 Marks]

- Ammonia has more atoms / more bonds per molecule (4 atoms vs 2 in H_2) [1]
- Hence NH_3 has more vibrational, rotational, and translational modes / more microstates to disperse quanta of energy [1].

(b) [3 Marks]

- $S^\circ_{\text{system}} = S^\circ(\text{products}) - S^\circ(\text{reactants}) = 2(192.3) - [191.6 + 3(130.6)]$ [1]
- $= 384.6 - [191.6 + 391.8] = 384.6 - 583.4$ [1]
- $= -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ (must include minus sign and correct value) [1].

(c) [4 Marks]

- (i) At 298 K: $S_{\text{surr}} = -(-92\,220 \text{ J mol}^{-1}) / 298 \text{ K} = +309.46 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]
- (ii) At 700 K: $S_{\text{surr}} = -(-92\,220 \text{ J mol}^{-1}) / 700 \text{ K} = +131.74 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]

Examiner Trap: Forgetting to multiply H by 1000 loses 1 mark per calculation!

(d) [4 Marks]

- At 298 K: $S_{\text{total}} = -198.8 + 309.46 = +110.66 \text{ J K}^{-1} \text{ mol}^{-1} \Rightarrow S_{\text{total}} > 0$, feasible [2]
- At 700 K: $S_{\text{total}} = -198.8 + 131.74 = -67.06 \text{ J K}^{-1} \text{ mol}^{-1} \Rightarrow S_{\text{total}} < 0$, non-feasible [2].

(e) [5 Marks]

- Crossover when $S_{\text{total}} = 0 \Rightarrow S_{\text{surr}} = -S_{\text{sys}} \Rightarrow -H/T = -S_{\text{sys}} \Rightarrow T = H / S_{\text{sys}}$ [1]
- $T = (-92\,220) / (-198.8) = 463.9 \text{ K}$ (approx 191 °C) [2]
- Compromise explanation: At 298 K / low T, reaction is feasible but rate is imperceptibly slow due to high activation energy of breaking NN triple bond [1]
- 700 K with iron catalyst provides acceptable rate of reaction; unreacted gases are recycled [1].

Question 32: Thermal Decomposition of Calcium Carbonate (18 Marks)

(a) [2 Marks]

- 1 mole of solid forms 1 mole of solid plus 1 mole of gas [1]
- Gases have vastly more disordered particles / higher microstate distribution of kinetic energy [1].

(b) [3 Marks]

- $S^\circ_{\text{system}} = S^\circ(\text{CaO}) + S^\circ(\text{CO}_2) - S^\circ(\text{CaCO}_3)$ [1]
- $= 38.2 + 213.6 - 92.9$ [1]
- $= +158.9 \text{ J K}^{-1} \text{ mol}^{-1}$ [1].

(c) [4 Marks]

- $S_{\text{surr}} = -(+178\,300) / 298 = -598.32 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $S_{\text{total}} = +158.9 + (-598.32) = -439.42 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]
- S_{total} is strongly negative, so the reaction is completely non-spontaneous at room temperature [1].

(d) [4 Marks]

- Feasibility when $S_{\text{total}} \geq 0 \Rightarrow T_{\text{min}} = H / S_{\text{system}}$ [1]
- $T_{\text{min}} = 178\,300 / 158.9 = 1122.1 \text{ K}$ [2]
- In °C: $1122.1 - 273.15 = 848.9 \text{ °C} \approx 849 \text{ °C}$ [1].

(e) [5 Marks]

- Mg^{2+} has a smaller ionic radius than Ca^{2+} (0.072 nm vs 0.100 nm) [1]
- Both have +2 charge, so Mg^{2+} has a much higher charge density [1]
- Mg^{2+} exerts greater polarising power on the large, polarisable CO_3^{2-} electron cloud [1]
- This weakens the C-O bond within carbonate, facilitating release of CO_2 [1]
- Lattice energy difference between MgO and MgCO_3 is more favourable than CaO and CaCO_3 [1].

Question 33: Dissolution Calorimetry & Thermodynamic Driving Forces (18 Marks)

(a) [4 Marks]

- Mass of solution = 50.0 g (from volume 50 cm^3 and density 1.0 g cm^{-3}) [1]
- $q = m c T = 50.0 \text{ g} \times 4.18 \text{ J g}^{-1} \text{ K}^{-1} \times 12.4 \text{ K} = 2591.6 \text{ J} = 2.5916 \text{ kJ}$ [1]

- Moles of $\text{NH}_4\text{NO}_3 = 8.00 / 80.0 = 0.100 \text{ mol}$ [1]
- $H_{\text{sol}} = +q / n = +2.5916 / 0.100 = +25.92 \text{ kJ mol}^{-1}$ (endothermic, must have + sign) [1].

(b) [4 Marks]

- $S_{\text{surr}} = -H / T = -25\,916 \text{ J mol}^{-1} / 298 \text{ K} = -86.97 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]
- $S_{\text{total}} = S_{\text{system}} + S_{\text{surr}} = +108.7 + (-86.97) = +21.73 \text{ J K}^{-1} \text{ mol}^{-1}$ [2].

(c) [3 Marks]

- Feasibility is dictated by S_{total} , NOT H alone [1]
- Although heat is taken from surroundings ($S_{\text{surr}} < 0$), the breakdown of the crystal lattice creates immense disorder ($S_{\text{sys}} = +108.7$) [1]
- Because S_{system} outweighs the negative $S_{\text{surroundings}}$, $S_{\text{total}} > 0$, satisfying the 2nd Law [1].

(d) [3 Marks]

- Dissolving releases free ions, but Cu^{2+} has high charge (+2) and relatively small radius [1]
- Water dipoles are strongly attracted to Cu^{2+} and SO_4^{2-} , forming rigid, highly ordered hydration shells [1]
- The loss of disorder from ordering of solvent water molecules exceeds the gain in disorder from crystal lattice breakdown [1].

(e) [4 Marks]

- Feasible when $S_{\text{total}} \geq 0 \Rightarrow S_{\text{sys}} - H/T \geq 0 \Rightarrow -156.0 - (-66\,500 / T) \geq 0$ [1]
- $66\,500 / T \geq 156.0 \Rightarrow T \leq 66\,500 / 156.0 = 426.28 \text{ K}$ [2]
- In $^{\circ}\text{C}$: $426.28 - 273.15 = 153.1^{\circ}\text{C}$ [1].
- Assumption: H and S_{system} do not change significantly with temperature.

Question 34: *Level of Response — Combustion Thermodynamics & Fuels (18 Marks)

(a) [5 Marks]

- (i) $S^{\circ}_{\text{system}} = [213.6 + 2(69.9)] - [186.2 + 2(205.0)] = 353.4 - 596.2 = -242.8 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]
- (ii) $S_{\text{surr}} = -(-890\,300) / 298 = +2987.58 \text{ J K}^{-1} \text{ mol}^{-1}$ [2]
- (iii) $S_{\text{total}} = -242.8 + 2987.58 = +2744.78 \text{ J K}^{-1} \text{ mol}^{-1}$ [1].

*(b) [6 Marks — Level of Response]

- Level 3 (5–6 marks): Correctly calculates ethanol $S_{\text{surr}} = +4588.26$ and $S_{\text{total}} = +4450.16 \text{ J K}^{-1} \text{ mol}^{-1}$. Explains that massive exothermic enthalpy releases heat into surroundings, generating huge positive S_{surr} that overwhelms negative S_{sys} . Explains that higher temperature reduces S_{surr} ($-H/T$), slightly decreasing S_{total} (though still huge positive). Evaluates bioethanol as renewable / potentially carbon neutral vs fossil methane.
- Level 2 (3–4 marks): Correct calculations with minor slip; explains dominance of $S_{\text{surroundings}}$; basic comparison of environmental impact.
- Level 1 (1–2 marks): Partial calculation; identifies that both are feasible.

(c) [4 Marks]

- Thermodynamic feasibility indicates the direction of spontaneous change ($S_{\text{total}} > 0$), but says nothing about reaction rate [1]
- C–H bonds in methane (413 kJ mol^{-1}) and O=O in oxygen (498 kJ mol^{-1}) require large energy to break [1]
- At 298 K, negligible fraction of molecules have energy $E \geq E_a$ (kinetic stability / inertness) [1]
- A spark or flame provides energy to overcome E_a , triggering self-sustaining reaction [1].

(d) [3 Marks]

- At boiling point (373 K), liquid and vapour are in dynamic equilibrium: $S_{\text{total}} = 0$ [1]
- $S_{\text{vap}} = H_{\text{vap}} / T_b = +40\,700 \text{ J mol}^{-1} / 373 \text{ K}$ [1]
- $= +109.12 \text{ J K}^{-1} \text{ mol}^{-1}$ (acceptable range: +109.0 to +109.2) [1].

Question 35: Synoptic N_2O_5 Decomposition — Kinetics & Thermodynamics (18 Marks)

(a) [4 Marks]

- $t_{1/2} = \ln 2 / k = 0.69315 / (3.46 \times 10^{-5} \text{ s}^{-1}) = 20\,033 \text{ s}$ [2]
- In hours: $20\,033 / 3600 = 5.56 \text{ hours}$ [1]
- Successive half-lives on a $[\text{N}_2\text{O}_5]$ vs time graph are constant, which is unique diagnostic for 1st order [1].

(b) [4 Marks]

- $\ln(k_2 / k_1) = \ln(4.87 \times 10^{-3} / 3.46 \times 10^{-5}) = \ln(140.75) = 4.9470$ [1]

- $(1/T_1 - 1/T_2) = (1/298 - 1/338) = 0.0033557 - 0.0029586 = 3.971 \times 10^{-4} \text{ K}^{-1} [1]$
- $E_a / R = 4.9470 / (3.971 \times 10^{-4}) = 12\,457.8 \text{ K} [1]$
- $E_a = 12\,457.8 \times 8.314 = 103\,574 \text{ J mol}^{-1} = +103.6 \text{ kJ mol}^{-1} [1]$.

(c) [3 Marks]

- $S^\circ_{\text{system}} = [4 \times S^\circ(\text{NO}_2) + S^\circ(\text{O}_2)] - 2 \times S^\circ(\text{N}_2\text{O}_5) = [4(240.0) + 205.0] - 2(356.0) [1]$
- $= [960.0 + 205.0] - 712.0 = 1165.0 - 712.0 [1]$
- $= +453.0 \text{ J K}^{-1} \text{ mol}^{-1} [1]$.

(d) [4 Marks]

- $S_{\text{surr}} = -H / T = -(+109\,500) / 298 = -367.45 \text{ J K}^{-1} \text{ mol}^{-1} [2]$
- $S_{\text{total}} = +453.0 + (-367.45) = +85.55 \text{ J K}^{-1} \text{ mol}^{-1} [1]$
- Since $S_{\text{total}} > 0$, reaction IS thermodynamically feasible at 298 K [1].

(e) [3 Marks]

- Kinetics: Higher T shifts Maxwell-Boltzmann curve to right, exponentially increasing fraction of molecules with $E \geq E_a$ (103.6 kJ mol⁻¹), multiplying reaction rate dramatically [2]
- Thermodynamics: Higher T decreases magnitude of negative S_{surr} ($-H/T$), making S_{total} even more positive (+140 J K⁻¹ mol⁻¹ at 350 K), shifting equilibrium further toward products [1].

Question 36: Steam Reforming of Methane & Industrial Hydrogen (15 Marks)

(a) [3 Marks]

- $S^\circ_{\text{system}} = [S^\circ(\text{CO}) + 3 \times S^\circ(\text{H}_2)] - [S^\circ(\text{CH}_4) + S^\circ(\text{H}_2\text{O})] [1]$
- $= [197.6 + 3(130.6)] - [186.2 + 188.7] = [197.6 + 391.8] - 374.9 = 589.4 - 374.9 [1]$
- $= +214.5 \text{ J K}^{-1} \text{ mol}^{-1} [1]$.

(b) [4 Marks]

- Feasible when $S_{\text{total}} \geq 0 \Rightarrow T \geq H / S_{\text{system}} [1]$
- $T_{\text{min}} = 206\,100 \text{ J mol}^{-1} / 214.5 \text{ J K}^{-1} \text{ mol}^{-1} = 960.84 \text{ K} [2]$
- In °C: $960.84 - 273.15 = 687.7 \text{ °C} \approx 688 \text{ °C} [1]$.

(c) [5 Marks]

- (i) Endothermic: Higher T increases S_{total} (less negative S_{surr}), substantially increasing K_p and equilibrium yield of H₂; also increases rate to reach equilibrium quickly [2]
- (ii) Reactants = 2 mol gas; Products = 4 mol gas. Increasing pressure shifts position of equilibrium to side with fewer moles of gas (to left), reducing yield of H₂ [2]
- (iii) High pressure increases collision frequency / rate of reaction, reduces reactor volume required (capital cost), and delivers H₂ at high pressure for downstream processing [1].

(d) [3 Marks]

- $S_{\text{surr}} = -(-41\,200) / 500 = +82.4 \text{ J K}^{-1} \text{ mol}^{-1} [1]$
- $S_{\text{total}} = -42.1 + 82.4 = +40.3 \text{ J K}^{-1} \text{ mol}^{-1} [1]$
- Reaction is exothermic: operating at lower temperature increases $S_{\text{surroundings}}$ and S_{total} , maximising conversion of poisonous CO to CO₂ and H₂ [1].

Question 37: Biophysical Thermodynamics of Protein Folding (15 Marks)

(a) [2 Marks]

- Unfolded protein exists as flexible random coil with billions of possible conformational states [1]
- Native folded state is a single, highly constrained 3D tertiary structure, drastically reducing conformational microstates ($S_{\text{chain}} < 0$) [1].

(b) [4 Marks]

- $S_{\text{surr}} = -(-280\,000 \text{ J mol}^{-1}) / 298 \text{ K} = +939.60 \text{ J K}^{-1} \text{ mol}^{-1} [2]$
- $S_{\text{total}} = -752.0 + 939.60 = +187.60 \text{ J K}^{-1} \text{ mol}^{-1} [1]$
- Since $S_{\text{total}} > 0$, the folded enzyme is thermodynamically stable and spontaneous at 298 K [1].

(c) [4 Marks]

- At melting temperature T_m , folded and unfolded states in equilibrium: $S_{\text{total}} = 0 [1]$
- $T_m = H^\circ_{\text{fold}} / S^\circ_{\text{system}} = (-280\,000) / (-752.0) = 372.34 \text{ K} [2]$
- In °C: $372.34 - 273.15 = 99.2 \text{ °C} [1]$.

(d) [5 Marks]

- In unfolded polypeptide, water molecules form highly ordered hydrogen-bonded 'clathrate cages' around exposed non-polar side chains, lowering water entropy [2]
 - Upon folding, non-polar side chains cluster into internal core, releasing ordered cage water molecules into bulk liquid [2]
 - The immense increase in solvent water entropy (positive S_{solvent}) compensates for the loss of polypeptide conformational entropy, driving spontaneous folding [1].
-